

วิธีการรันและผลลัพธ์

1.เปิด Docker containers ทั้งหมด

- docker compose up -d

```
boomn@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DADS6002/code (main)
$ docker compose up -d
[+] up 12/12
✓ Network code_default Created
✓ Container hive-metastore-postgresql Started
✓ Container historyserver Started
✓ Container resourcemanager Started
✓ Container namenode Started
✓ Container spark-master Started
✓ Container datanode Started
✓ Container nodemanager Started
✓ Container hive-metastore Started
✓ Container spark-worker Started
✓ Container hive-server Started
✓ Container hue Started

what's next:
  Filter, search, and stream logs from all your Compose services
  in one place with Docker Desktop's Logs view. docker-desktop://dashboard/logs?appId=code

boomn@DESKTOP-IU0AVN0 MINGW64 ~/OneDrive/Documents/GitHub/DADS6002/code (main)
$
```

2. รัน Spark Job

- MSYS_NO_PATHCONV=1 ./spark.sh

Spark RDD Output

```
sc@msys64:~/code$ ./spark.sh
26/06/02 15:25:00 INFO SparkContext: Running Spark version 3.2.1
26/06/02 15:25:00 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/06/02 15:25:00 INFO ResourceUtils: No custom resources configured for spark.driver.
26/06/02 15:25:00 INFO SparkContext: Submitted application: Assignment2.AverageIncome.RDD
26/06/02 15:25:00 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, s
cript: , vendor: , offheap -> name: offheap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/06/02 15:25:00 INFO ResourceProfile: Limiting resource is cpu
26/06/02 15:25:00 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/06/02 15:25:00 INFO SecurityManager: Changing view acls to: root
26/06/02 15:25:00 INFO SecurityManager: Changing modify acls to: root
26/06/02 15:25:00 INFO SecurityManager: Changing view acls groups to:
26/06/02 15:25:00 INFO SecurityManager: Changing modify acls groups to:
26/06/02 15:25:00 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(root); groups with view permissions: Set(); users with
modify permissions: Set(root); groups with modify permissions: Set()
26/06/02 15:25:00 INFO SparkEnv: Registering MapOutputTracker
26/06/02 15:25:00 INFO SparkEnv: Registering BlockManagerMaster
26/06/02 15:25:00 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/06/02 15:25:00 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, 7f33a3cd00cc, 35693, None)
26/06/02 15:25:00 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/06/02 15:25:00 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-f0a2d10e-1ac1-4a5c-ba80-f19d81145e05
26/06/02 15:25:00 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
26/06/02 15:25:00 INFO SparkContext: Registering OutputCommitCoordinator
26/06/02 15:25:01 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/06/02 15:25:01 INFO SparkUI: Bound SparkUI to 0.0.0.0, and started at http://7f33a3cd00cc:4040
26/06/02 15:25:01 INFO Executor: Starting executor ID driver on host 7f33a3cd00cc
26/06/02 15:25:01 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 35693.
26/06/02 15:25:01 INFO NettyBlockTransferService: Server created on 7f33a3cd00cc:35693
26/06/02 15:25:01 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/06/02 15:25:01 INFO BlockManager: Registered BlockManager BlockManagerId(driver, 7f33a3cd00cc, 35693, None)
26/06/02 15:25:01 INFO BlockManagerMasterEndpoint: Registering block manager 7f33a3cd00cc:35693 with 366.3 MiB RAM, BlockManagerId(driver, 7f33a3cd00cc, 35693, None)
26/06/02 15:25:01 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 7f33a3cd00cc, 35693, None)
26/06/02 15:25:01 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 7f33a3cd00cc, 35693, None)
26/06/02 15:25:01 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/06/02 15:25:01 INFO SharedState: Warehouse path is 'file:/spark-warehouse'.

Part 1: Using Spark RDD
=====
District 1: Average Income = 163214.29
District 2: Average Income = 103600.00
District 3: Average Income = 59300.00
District 4: Average Income = 30571.43
District 5: Average Income = 20888.89
```

Spark SQL Output

```
What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image - docker debug spark-master
  Learn more at https://docs.docker.com/go/debug-cli/

Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
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26/06/02 15:25:00 INFO ResourceUtils: No custom resources configured for spark.driver.
26/06/02 15:25:00 INFO ResourceUtils:
26/06/02 15:25:00 INFO SparkContext: Submitted application: Assignment2.AverageIncome.Sql
26/06/02 15:25:00 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , of
fheap -> name: offheap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
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26/06/02 15:25:00 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(root); groups with view permissions: Set(); users with modify permissions:
Set(root); groups with modify permissions: Set()
26/06/02 15:25:00 INFO SparkEnv: Registering MapOutputTracker
26/06/02 15:25:00 INFO SparkEnv: Registering BlockManagerMaster
26/06/02 15:25:00 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/06/02 15:25:00 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, 7f33a3cd00cc, 42345, None)
26/06/02 15:25:00 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/06/02 15:25:00 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-9402c4cb-35d4-4315-a073-f66efb2db07
26/06/02 15:25:00 INFO MemoryStore: MemoryStore started with capacity 366.3 MiB
26/06/02 15:25:00 INFO SparkContext: Registering OutputCommitCoordinator
26/06/02 15:25:00 INFO SparkUI: Bound SparkUI to 0.0.0.0, and started at http://7f33a3cd00cc:4040
26/06/02 15:25:00 INFO Executor: Starting executor ID driver on host 7f33a3cd00cc
26/06/02 15:25:00 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 42345.
26/06/02 15:25:00 INFO NettyBlockTransferService: Server created on 7f33a3cd00cc:42345
26/06/02 15:25:00 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/06/02 15:25:00 INFO BlockManager: Registered BlockManager BlockManagerId(driver, 7f33a3cd00cc, 42345, None)
26/06/02 15:25:00 INFO BlockManagerMasterEndpoint: Registering block manager 7f33a3cd00cc:42345 with 366.3 MiB RAM, BlockManagerId(driver, 7f33a3cd00cc, 42345, None)
26/06/02 15:25:00 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, 7f33a3cd00cc, 42345, None)
26/06/02 15:25:00 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, 7f33a3cd00cc, 42345, None)
26/06/02 15:25:00 INFO SharedState: Setting hive.metastore.warehouse.dir ('null') to the value of spark.sql.warehouse.dir.
26/06/02 15:25:00 INFO SharedState: Warehouse path is 'file:/spark-warehouse'.

Part 2: Using Spark SQL
=====
[district_id]    [average_income]
-----
1| 163214.2857142857|
2|           103600.0|
3|           59300.0|
4| 30571.428571428572|
5|  20888.888888888889|

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```